Complete Beginner's Guide: Running the Bhulekh RoR Scraper

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https://github.com/s-deba-sys/bhulekh-data-scrapper

This guide provides step-by-step instructions to set up and run the scraper on a Windows computer, even if you have no prior experience with coding, Python, or PowerShell.

Part 1: One-Time Setup on Your Computer

You only need to complete these steps once.

Step 1: Install Python

1. Visit the official Python download page: python.org/downloads.
2. Click the yellow button that says **Download Python**.
3. Once downloaded, double-click the file to start the installer.
4. **CRITICAL STEP:** On the very first installation screen, check the box at the bottom that says **"Add python.exe to PATH"**.

If you do not check this box, Windows will not recognize Python commands.

5. Click **Install Now** and wait for the installation to finish.

Step 2: Install Required Helper Libraries

1. Press the **Windows Key** on your keyboard, type powershell, and press **Enter**.
2. A blue or black window (**Windows PowerShell**) will appear.
3. Copy and paste the following command into PowerShell, then press **Enter**:

   ```
   pip install pandas beautifulsoup4 openpyxl requests
   ```

4. Wait until the download and installation finish (it will show Successfully installed...).
5. Close the PowerShell window.

Part 2: Preparing Your Files

Create a dedicated folder on your computer to keep everything organized.

Step 1: Create a Working Folder

1. Go to your **Desktop**.
2. Right-click an empty space, choose **New** → **Folder**, and name it: BhulekhScraper

Step 2: Prepare Your Input File (data.xlsx)

1. Open Microsoft Excel.
2. Arrange your columns in the exact order shown below:

Column A	Column B	Column C	Column D	Column E	Column F	Column G
District Name	**District Code**	**Tahasil Name**	**Tahasil Code**	**Village Name**	**Village Code**	**Khata No**
Puri	11	Puri	4	Samang	185	24
Puri	11	Gop	10	Adhana	44	200

3. Click **File** → **Save As**.
4. Choose your new **BhulekhScraper** folder on the Desktop.
5. Save the file with the exact name: data.xlsx

Step 3: Save the Python Script (scrape_bhulekh.py)

1. Press the **Windows Key**, type Notepad, and press **Enter**.
2. Paste the Python code from the section below into Notepad.
3. Click **File** → **Save As**.
4. In the Save dialog:
    - Navigate to your **BhulekhScraper** folder on the Desktop.
    - Change **Save as type** from Text Documents (*.txt) to **All Files (*.*)**.
    - Name the file: scrape_bhulekh.py
5. Click **Save**.

Python Code to Paste into Notepad:

```
import os

import re

import time

from bs4 import BeautifulSoup

import openpyxl

from openpyxl.styles import Alignment, Border, Font, PatternFill,
Side

from openpyxl.utils import get_column_letter

import pandas as pd
```

```
import requests

INPUT_FILE = "data.xlsx"
OUTPUT_FILE = "bhulekh_scraped_output.xlsx"

HEADERS = {
    "User-Agent": (
        "Mozilla/5.0 (Windows NT 10.0; Win64; x64) "
        "AppleWebKit/537.36 (KHTML, like Gecko) "
        "Chrome/124.0.0.0 Safari/537.36"
    )
}


def get_text_by_id(soup, element_id):
    elem = soup.find(id=element_id)
    return elem.text.strip() if elem else ""


def clean_plot_number(raw_text):
    if not raw_text:
        return ""
    match = re.match(r"^\s*([\d/\-]+)", str(raw_text))
    return match.group(1) if match else raw_text.strip()


def parse_khata_html(html_content, url=""):
    soup = BeautifulSoup(html_content, "html.parser")
    page_text = soup.get_text()
```

```
    if "ପରିଶିଷ୍ଟ - କ" in page_text:
        khata_type = "Consolidation"
    elif "Schedule I Form No.39-A" in page_text:
        khata_type = "Settlement"
    else:
        khata_type = "Unknown"

    mouja = get_text_by_id(soup, "gvfront_ctl02_lblMouja")
    tehsil = get_text_by_id(soup, "gvfront_ctl02_lblTehsil")
    thana = get_text_by_id(soup, "gvfront_ctl02_lblThana")
    district = get_text_by_id(soup, "gvfront_ctl02_lblDist")
    khatiyan_no = get_text_by_id(soup,
"gvfront_ctl02_lblKhatiyanslNo")
    tenant_name = get_text_by_id(soup, "gvfront_ctl02_lblName")
    landlord_name = get_text_by_id(soup,
"gvfront_ctl02_lblLandlordName")
    status = get_text_by_id(soup, "gvfront_ctl02_lblStatua")
    publish_date = get_text_by_id(soup,
"gvfront_ctl02_lblLastPublishDate")

    plot_list = []
    total_khata_acres = 0.0

    ror_table = soup.find("table", id="gvRorBack")
    if ror_table:
        rows = ror_table.find_all("tr")
        for row in rows:
            col_8_elem =
row.find(id=re.compile(r"gvRorBack_ctl\d+_lblPlotNo"))
            plot_ci_elem = row.find(
                id=re.compile(r"gvRorBack_ctl\d+_lblPlotci")
            )
```

```
            plot_cni_elem = row.find(
                id=re.compile(r"gvRorBack_ctl\d+_lblPlotcni")
            )

            if col_8_elem or plot_ci_elem or plot_cni_elem:
                chaka_no = ""
                actual_plot_no = ""
                ltype = ""

                col_8_text = col_8_elem.text.strip() if col_8_elem
else ""
                plot_ci_text = (
                    plot_ci_elem.text.strip() if plot_ci_elem else
""
                )
                plot_cni_text = (
                    plot_cni_elem.text.strip() if plot_cni_elem else
""
                )

                if khata_type == "Consolidation":
                    if plot_ci_text:
                        actual_plot_no =
clean_plot_number(plot_ci_text)
                        chaka_no = clean_plot_number(col_8_text)
                        ltype_elem = row.find(

id=re.compile(r"gvRorBack_ctl\d+_lbllType")
                        )
                        ltype = ltype_elem.text.strip() if
ltype_elem else ""
                    elif plot_cni_text:
```

```
                        actual_plot_no =
clean_plot_number(plot_cni_text)
                        chaka_no = ""
                        cni_elem = row.find(

id=re.compile(r"gvRorBack_ctl\d+_lblCNItype")
                        )
                        ltype = cni_elem.text.strip() if cni_elem
else ""
                else:
                    actual_plot_no = clean_plot_number(col_8_text)
                    chaka_no = ""
                    ltype_elem = row.find(
                        id=re.compile(r"gvRorBack_ctl\d+_lbllType")
                    )
                    ltype = ltype_elem.text.strip() if ltype_elem
else ""

                if actual_plot_no:
                    acre_elem = row.find(
                        id=re.compile(r"gvRorBack_ctl\d+_lblAcre")
                    )
                    dec_elem = row.find(

id=re.compile(r"gvRorBack_ctl\d+_lblDecimil")
                    )

                    acre_str = acre_elem.text.strip() if acre_elem
else "0"
                    dec_str = dec_elem.text.strip() if dec_elem else
"0"

                    try:
```

```
                        x_acre = float(acre_str) if acre_str else
0.0
                    except ValueError:
                        x_acre = 0.0

                    try:
                        yyyy_dec = float(dec_str) if dec_str else
0.0
                    except ValueError:
                        yyyy_dec = 0.0

                    calculated_acres = round(x_acre + (yyyy_dec /
10000.0), 4)
                    total_khata_acres += calculated_acres

                    entry_parts = [f"Plot: {actual_plot_no}"]
                    if chaka_no:
                        entry_parts.append(f"Chaka: {chaka_no}")
                    if ltype:
                        entry_parts.append(f"Kisam: {ltype}")
                    entry_parts.append(f"Area: {calculated_acres}
Acres")

                    plot_list.append(f"[{' | '.join(entry_parts)}
]")

    total_plots = len(plot_list)
    plots_formatted = (
        ",\n".join(plot_list) if total_plots > 0 else "No Plots
Available"
    )
```

```
    return {
        "URL": url,
        "Khata_Type": khata_type,
        "Khatiyan_No": khatiyan_no,
        "Mouja": mouja,
        "Tehsil": tehsil,
        "Thana": thana,
        "District": district,
        "Tenant_Name": tenant_name,
        "Landlord_Name": landlord_name,
        "Status": status,
        "Publish_Date": publish_date,
        "Total_Plots_Count": total_plots,
        "Total_Khata_Area_Acres": round(total_khata_acres, 4),
        "Plot_Details": plots_formatted,
    }


def main():
    print("=" * 65)
    print("Bhulekh Odisha Auto-URL Generator & RoR Scraper")
    print("=" * 65)

    if not os.path.exists(INPUT_FILE):
        print(f"[ERROR] '{INPUT_FILE}' not found in current
folder.")
        return

    df_input = pd.read_excel(INPUT_FILE)
    print(f"Loaded '{INPUT_FILE}' with {len(df_input)} rows.\n")
```

```
    col_dist_code = df_input.columns[1]
    col_teh_code = df_input.columns[3]
    col_vill_code = df_input.columns[5]
    col_khata_no = (
        df_input.columns[6] if len(df_input.columns) > 6 else
"KhataNo"
    )

    session = requests.Session()
    session.headers.update(HEADERS)
    results = []

    for idx, row in df_input.iterrows():
        try:
            dist_code = int(row[col_dist_code])
            teh_code = int(row[col_teh_code])
            vill_code = int(row[col_vill_code])
            khata_val = row.get(col_khata_no, 1)
            khata_no = int(khata_val) if pd.notna(khata_val) else 1

            target_url = (
                f"http://bhulekh.ori.nic.in/ViewRoR.aspx?"

f"DistCode={dist_code}&TehCode={teh_code}&VillCode={vill_code}&Khata
No={khata_no}&type=front"
            )
        except Exception as e:
            print(f"[{idx+1}/{len(df_input)}] Error reading row
codes: {e}")
            continue

        print(f"[{idx+1}/{len(df_input)}] Scraping: {target_url}")
```

```
        try:
            response = session.get(target_url, timeout=20)
            if response.status_code == 200:
                parsed = parse_khata_html(response.text,
url=target_url)
                results.append(parsed)
                print(
                    f"   -> Success: [{parsed['Khata_Type']}] "
                    f"Khata: {parsed['Khatiyan_No']}, Plots:
{parsed['Total_Plots_Count']}"
                )
            else:
                results.append(
                    {"URL": target_url, "Khata_Type": f"HTTP
{response.status_code}"}
                )
        except Exception as e:
            print(f"   -> Timeout / Network Error: {e}")
            results.append({"URL": target_url, "Khata_Type": "Error
/ Timeout"})

        time.sleep(0.1)

    print(f"\nWriting output to '{OUTPUT_FILE}'...")
    output_df = pd.DataFrame(results)

    with pd.ExcelWriter(OUTPUT_FILE, engine="openpyxl") as writer:
        output_df.to_excel(writer, index=False,
sheet_name="Scraped_Data")
        wb = writer.book
        ws = writer.sheets["Scraped_Data"]
```

```
        header_fill = PatternFill(
            start_color="1F497D", end_color="1F497D",
fill_type="solid"
        )
        header_font = Font(name="Calibri", size=11, bold=True,
color="FFFFFF")
        thin_border = Border(
            left=Side(style="thin", color="D9D9D9"),
            right=Side(style="thin", color="D9D9D9"),
            top=Side(style="thin", color="D9D9D9"),
            bottom=Side(style="thin", color="D9D9D9"),
        )

        for cell in ws[1]:
            cell.fill = header_fill
            cell.font = header_font
            cell.alignment = Alignment(
                horizontal="center", vertical="center",
wrap_text=True
            )

        for row in ws.iter_rows(
            min_row=2, max_row=ws.max_row, min_col=1,
max_col=ws.max_column
        ):
            for cell in row:
                cell.border = thin_border
                if cell.column == 14:
                    cell.alignment = Alignment(vertical="top",
wrap_text=True)
                else:
```

```
                    cell.alignment = Alignment(vertical="top")


        for col in ws.columns:
            max_len = max(len(str(cell.value or "")) for cell in
col)
            col_letter = get_column_letter(col[0].column)
            if col_letter == "N":
                ws.column_dimensions[col_letter].width = 65
            else:
                ws.column_dimensions[col_letter].width = min(
                    max(max_len + 3, 12), 35
                )


    print("=" * 65)
    print(f"Scraping completed! Results saved to '{OUTPUT_FILE}'.")
    print("=" * 65)



if __name__ == "__main__":
    main()
```

Part 3: Running the Program

1. Open your **BhulekhScraper** folder on your Desktop.
2. In the folder, click on the **Address Bar** at the top (where the folder path is displayed).
3. Type powershell directly into the address bar and press **Enter**.
    - *This opens PowerShell pointing directly at your folder.*
4. In the PowerShell window, type:

    ```
    python scrape_bhulekh.py
    ```

5. Press **Enter**.

Part 4: Reviewing Your Results

- As the script runs, you will see real-time status messages showing which URLs are being generated and processed.

- Once the prompt says **Scraping completed!**, open your BhulekhScraper folder.
- A new file named **bhulekh_scraped_output.xlsx** will be ready with all metadata, cleaned plot numbers, and calculated land areas.

Troubleshooting Quick-Reference

- **'python' is not recognized...**: Reinstall Python from Part 1 and ensure you check **"Add python.exe to PATH"**.
- **[ERROR] 'data.xlsx' not found...**: Make sure your Excel file is named data.xlsx and is located in the exact same folder as scrape_bhulekh.py.
- **No module named 'pandas' / 'requests'**: Re-run the installation command from Part 1, Step 2 in PowerShell.